

★ WEEK 5 · PART 1 of 3

ARCH: Modeling Conditional Variance

You Can't Forecast Direction — But You Can Forecast Risk

Conditional vs unconditional variance · ARCH(q) · the clustering engine

Weeks 3–4 modeled the mean and found returns nearly unpredictable. The structure lives in the variance. Part 1 introduces the idea that finally captures it: a variance that changes over time.

1

PART 1 • ARCH

When Variance Isn't Constant

The conditional-variance idea, the ARCH model that first captured volatility clustering, and the fat tails and diagnostics that come with it.

The Economist Who Modeled Fear — Part 1

How a question about inflation changed risk forever

Robert Engle asked a simple question: is the UNCERTAINTY of inflation itself changing over time? Standard models assumed a fixed variance — and he suspected they were wrong.

The setting

- Late 1970s inflation was volatile, then calm, then volatile again.
- Every model of the day assumed a single, constant variance.
- But the uncertainty itself seemed to cluster in time.

The insight

- What if variance depends on the recent past, like the mean can?
- Then risk is forecastable even when direction is not.
- That question became the ARCH model.

The Economist Who Modeled Fear — Part 2

ARCH is born, and the lesson

Engle's 1982 ARCH model let the variance evolve with recent shocks. It reshaped risk management and earned him the 2003 Nobel — shared with Granger, of cointegration fame.

What it changed

- Volatility became something you could estimate and forecast.
- Value-at-Risk, option pricing, and risk desks were rebuilt on it.
- It is now the default lens for financial risk.

The lesson for us

- Give up on forecasting direction; forecast risk instead.
- Model the variance as its own dynamic process.
- This is where the predictable structure in returns lives.

From the Mean to the Variance

Three ideas that set up this week

1

The mean of returns is nearly unpredictable

Weeks 3–4 showed ARIMA finds almost no linear structure in returns. Their ACF is essentially zero — direction bets have no statistical edge.

2

But their VARIANCE is highly predictable

Squared returns are strongly autocorrelated: big moves cluster. That is the stylized fact of Week 1, and the thing ARCH was built to model.

3

So we switch targets: model the variance

Keep the mean simple, and put the dynamics in the variance. That is the entire ARCH/GARCH program.

Week 5 in one line: you cannot forecast the return, but you can forecast its risk.

Why Constant Variance Fails

Markets are not equally risky every day

Every model so far assumed one fixed variance for all time. But real markets alternate between calm and crisis — a single number cannot describe both a quiet Tuesday and a crash.

What we observe

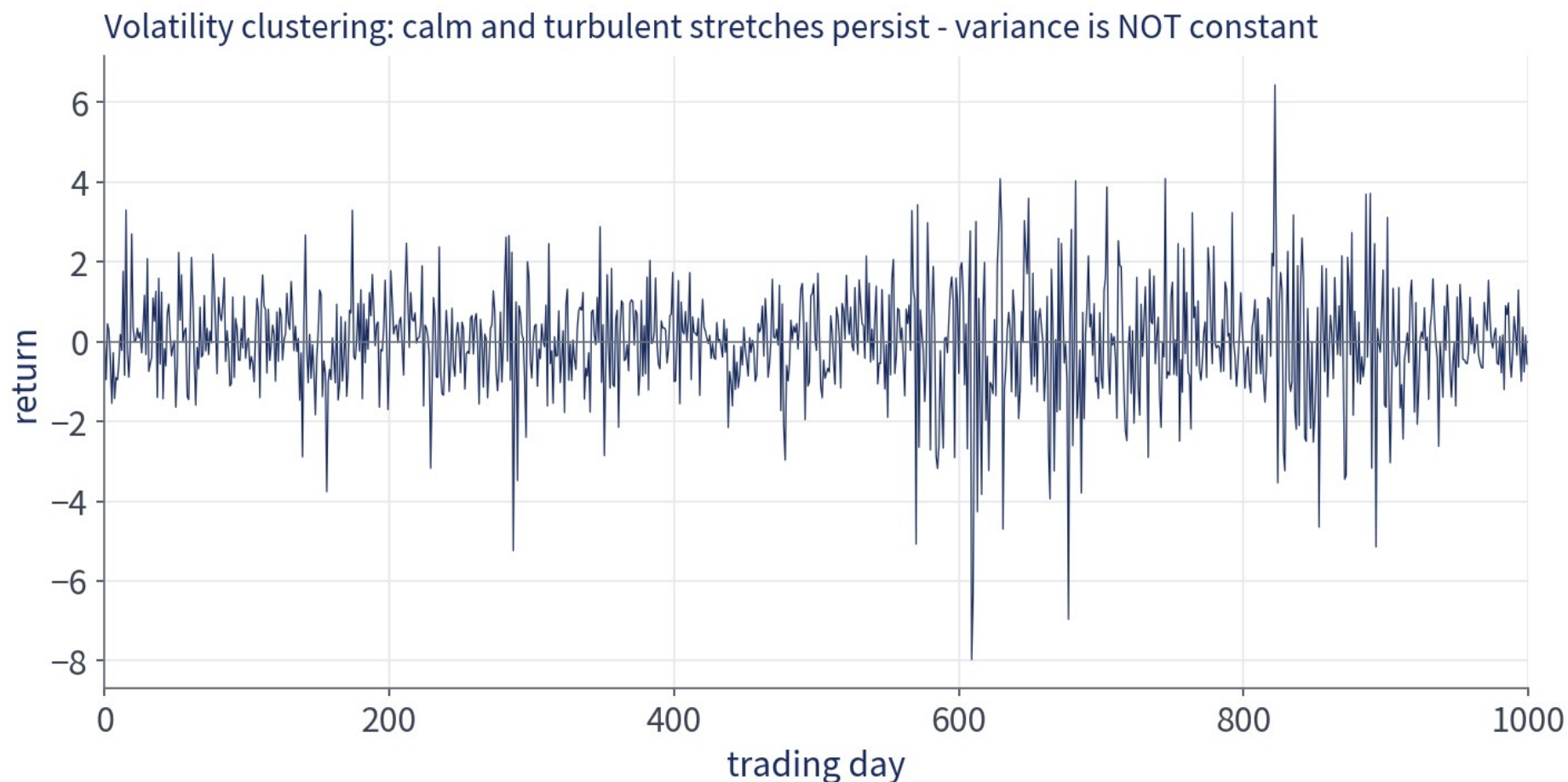
- Quiet weeks of tiny moves, then bursts of violent ones.
- The turbulence persists — clustering, not random spikes.
- A constant-variance model is right on average, wrong every day.

Why it matters

- Risk limits set on average variance blow up in a crisis.
- Option prices and margins need TODAY's risk, not the average.
- We need a variance that adapts — day by day.

Volatility Clustering, One More Time

Calm and turbulent stretches each persist — variance is plainly not constant. This is the pattern (Week 1) that a constant-variance model cannot represent.



Conditional vs Unconditional Variance

The single most important idea this week

$$\underbrace{\text{Var}(r_t) = \sigma^2}_{\text{unconditional: fixed}} \quad \underbrace{\text{Var}(r_t \mid \mathcal{F}_{t-1}) = \sigma_t^2}_{\text{conditional: time-varying}}$$

Unconditional variance

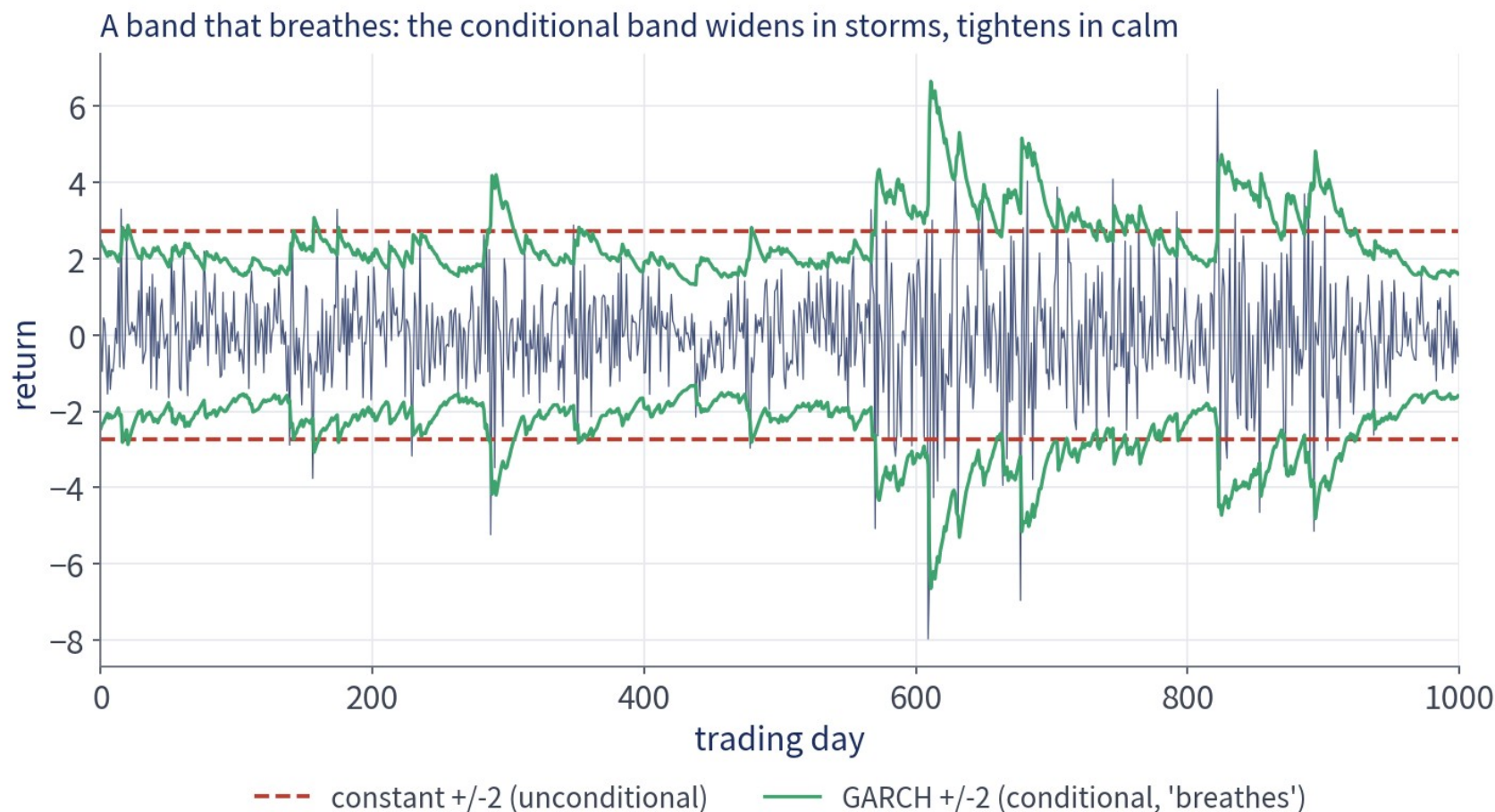
- The long-run, overall variance — one fixed number.
- It ignores what just happened in the market.
- Useful as an average, useless for today's risk.

Conditional variance

- The variance GIVEN the recent past — it changes each day.
- It is large after big shocks, small after calm.
- This time-varying number is what ARCH models.

A Band That Breathes With the Market

The constant (unconditional) band is the same every day. The conditional band widens in storms and tightens in calm — this is the risk number you actually want.



Two Equations: Mean and Variance

ARCH models the second one

A volatility model has two parts: a MEAN equation (often just a constant, since returns are unpredictable) and a VARIANCE equation that lets today's risk depend on the past.

The mean equation

- Usually as simple as 'return = constant + shock'.
- We already know there is little to model here (Weeks 3–4).
- Sometimes a small ARMA, but often nothing at all.

The variance equation

- This is where ARCH/GARCH does its work.
- It makes the shock's SIZE depend on recent shocks.
- All the modeling effort now goes here.

Separating Size From Direction

Every return is a size times a direction

$$r_t = \mu + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t, \quad z_t \stackrel{iid}{\sim} (0, 1)$$

Reading it

- The shock splits into a time-varying size and a random sign.
- σ_t is the conditional standard deviation — the 'size'.
- z_t is a standardized draw with mean 0 and variance 1 — the 'direction'.

Why this split

- It isolates exactly the piece ARCH models: σ_t .
- The direction z_t stays unpredictable (efficient markets).
- Model the size; leave the sign to chance.

The ARCH(1) Model

Variance driven by the last squared shock

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 \quad (\omega > 0, \alpha \geq 0) \quad \text{ARCH(1)}$$

How to read it

- Today's variance is a baseline ω plus a reaction to the last shock.
- A big shock yesterday (large squared value) $\rightarrow$ big variance today.
- The parameter α sets how strongly shocks feed into variance.

Why squared

- Squaring makes both up and down shocks raise variance.
- Only the SIZE of the shock matters, not its sign.
- That is what produces clustering of large moves.

The Engine of Clustering

How one shock echoes into tomorrow's risk

A large shock today enlarges tomorrow's variance, which makes another large shock more likely, which enlarges the day-after variance — a self-reinforcing loop that produces clusters.

The feedback

- Big shock $\rightarrow$ higher next variance $\rightarrow$ likely another big shock.
- Calm $\rightarrow$ lower next variance $\rightarrow$ likely more calm.
- Turbulence and tranquility both become sticky.

The balance

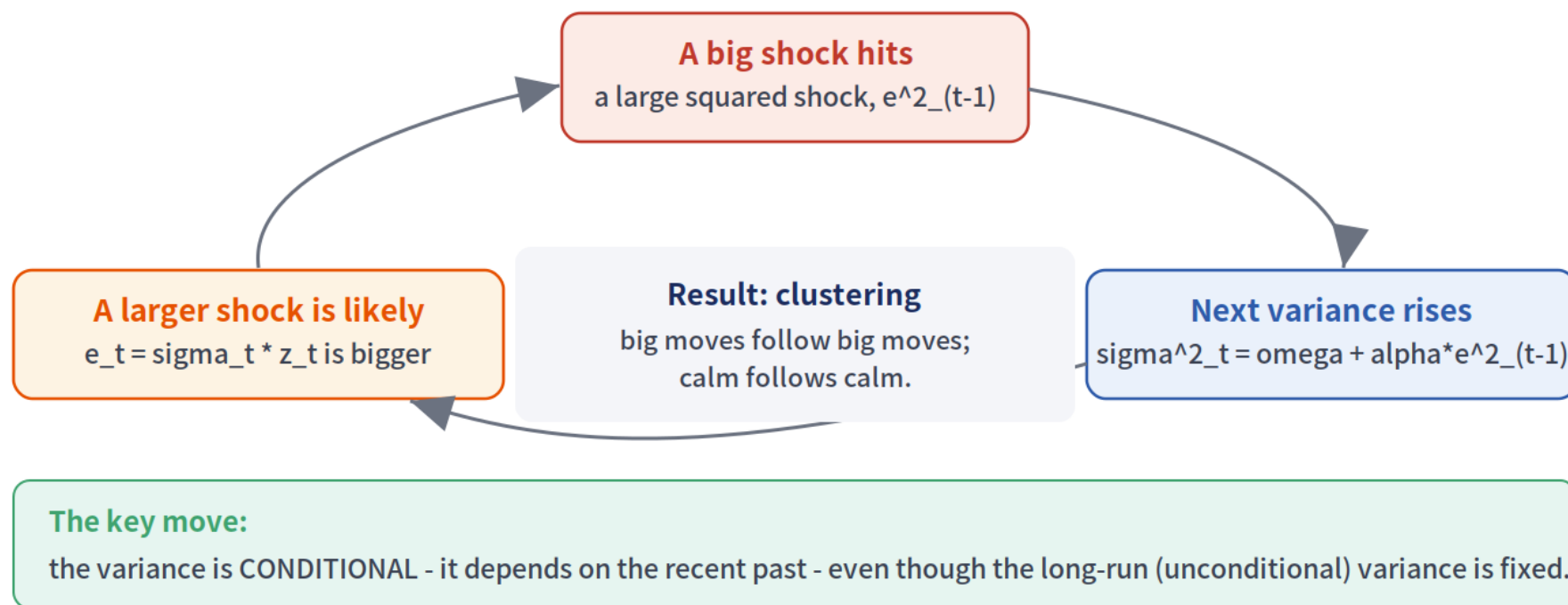
- ω anchors the long-run level of variance.
- α controls how twitchy the variance is to shocks.
- Too large an α and the variance explodes — bounds apply.

The Clustering Feedback Loop

ARCH ties today's variance to yesterday's shock. That single link turns isolated shocks into persistent clusters — while the long-run variance stays fixed.

The Engine of Volatility Clustering

ARCH makes today's variance depend on yesterday's shock - a self-reinforcing loop.



The ARCH(q) Model

Let the last q shocks all have a say

$$\sigma_t^2 = \omega + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2 \quad \text{ARCH}(q)$$

The extension

- Variance depends on the last q squared shocks, each weighted.
- q is the memory: how many days of shocks still matter.
- ARCH(1) is the special case $q = 1$.

The practical snag

- Real volatility has long memory — clusters last weeks.
- Capturing that with ARCH needs a large q (many parameters).
- That inefficiency is exactly what GARCH will fix (Part 2).

Unconditional Variance and Stationarity

When is an ARCH model well-behaved?

$$\mathbb{E}[\sigma_t^2] = \frac{\omega}{1 - \sum_i \alpha_i} \quad (\text{finite iff } \sum_i \alpha_i < 1)$$

The long-run variance

- Averaged over all time, the variance settles to $\omega/(1 - \sum \alpha)$.
- This is the unconditional variance the model reverts toward.
- The conditional variance orbits around this level.

The stationarity condition

- The α 's must sum to less than 1 for a finite variance.
- If they sum to 1 or more, variance grows without bound.
- So the reaction coefficients are always constrained.

Fat Tails, for Free

ARCH generates leptokurtosis by itself

Even with normal shocks z_t , mixing calm and turbulent periods produces returns with fatter tails than a normal — ARCH reproduces a stylized fact of Week 1 automatically.

Why it happens

- Some days draw from a wide variance, others from a narrow one.
- This mixture of variances has heavier tails than any single one.
- So extreme returns arise more often than a normal predicts.

The catch

- The tails ARCH makes are still not fat ENOUGH for real data.
- Real returns are heavier-tailed than plain ARCH implies.
- The fix is heavier-tailed shocks (Student-t) — Part 2.

ARCH and the Stylized Facts

One model, several facts at once

Facts it reproduces

- Volatility clustering — its whole reason for being.
- Fat tails — from the mixture of variances.
- Uncorrelated returns with correlated squared returns.

Why that is compelling

- A single mechanism explains several separate observations.
- That is the mark of a genuinely good model.
- It ties Week 1's facts directly to a workable equation.

Testing for ARCH Effects

Is there volatility to model at all?

ARCH-LM test: regress ε_t^2 on $\varepsilon_{t-1}^2, \dots, \varepsilon_{t-q}^2$; H_0 : no ARCH

How to check

- Look at the ACF of SQUARED returns — spikes mean ARCH effects.
- Formally, run the ARCH-LM test on the squared residuals.
- A small p-value says: yes, the variance has structure.

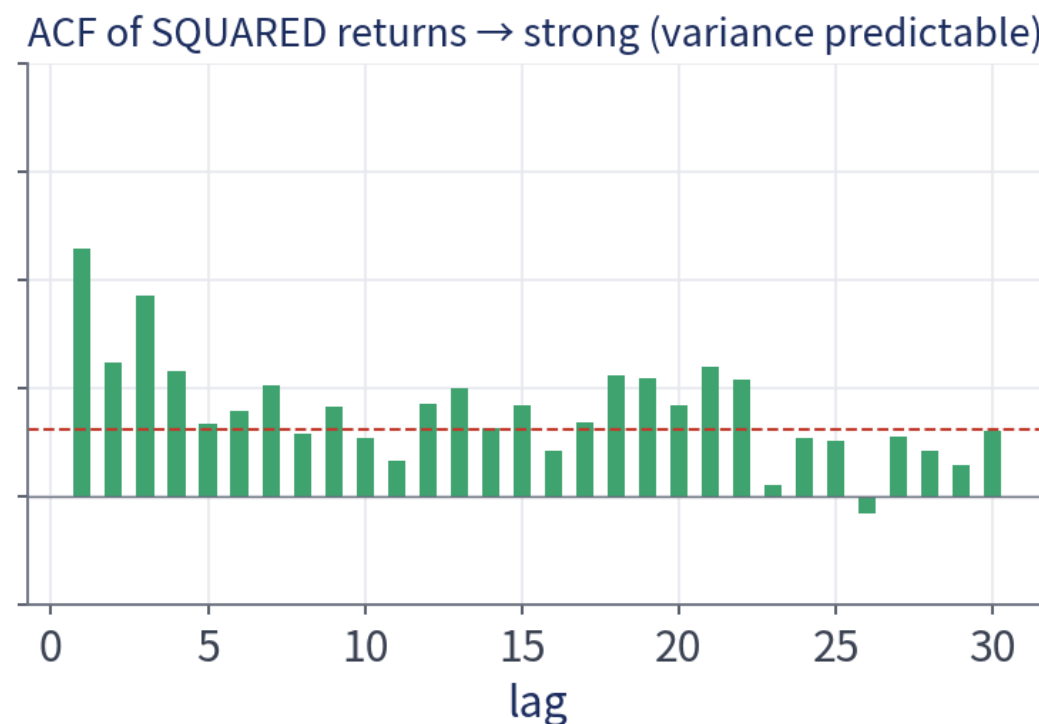
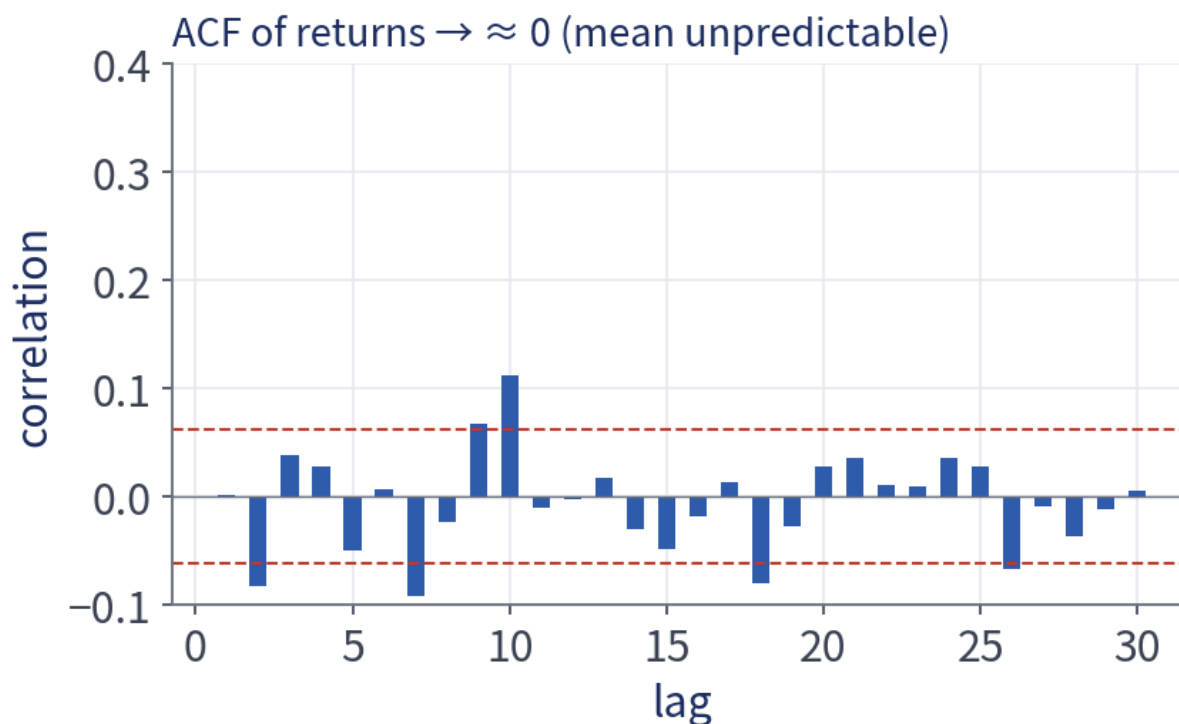
When to apply it

- Always test before fitting — confirm there is clustering.
- Also test the RESIDUALS after fitting your mean model.
- Leftover ARCH effects mean you need a volatility model.

The ARCH Signature

Returns are uncorrelated (left), yet squared returns are strongly autocorrelated (right). That gap is the fingerprint of ARCH effects — predictable variance.

The ARCH signature: squared returns are autocorrelated even when returns are not



Why ARCH Alone Isn't Enough

A great idea that needs one more step

ARCH captures clustering, but to match volatility's long memory it needs many lags and many parameters. GARCH will fix this with one elegant addition — memory of past variance.

The problem

- Real volatility clusters persist for weeks.
- Pure ARCH needs a high q to reach that far back.
- Many parameters mean noisy, over-fit estimates.

The preview

- GARCH adds a term for YESTERDAY's variance.
- One extra parameter captures long memory compactly.
- GARCH(1,1) does what ARCH(10+) struggles to — next part.

Building an ARCH Model

From returns to a fitted variance

1 • Mean

fit & remove

Fit a simple mean model;
keep the residuals.

2 • Test

ARCH-LM

Confirm the squared residuals show clustering.

3 • Specify

pick q

Choose the ARCH order from the squared-return ACF.

4 • Estimate

fit variance

Fit the ARCH by maximum likelihood.

5 • Check

standardize

Standardized residuals should be clean, no ARCH left.

The same discipline as Weeks 3–4, now applied to the variance instead of the mean.

Detect and Fit ARCH

Direct Claude Code from clustering to a variance model

GOAL Confirm ARCH effects in real returns, fit an ARCH model, and visualize the fitted conditional variance.

[INPUT] Prompt · Claude Code

Write and run `arch_lab.py` that:

1. downloads daily returns for a ticker and plots the ACF of returns vs squared returns;
2. runs the ARCH-LM test and reports whether ARCH effects are present;
3. fits an ARCH(q) with the 'arch' library (try $q = 1$ and a few larger q);
4. overlays the fitted conditional ± 2 -sigma band on the returns.

Tell me if the standardized residuals still show any ARCH structure.

How to work with it

- Expect flat return ACF, strong squared-return ACF.
- The band should widen in turbulent stretches.
- Larger q helps — but watch the parameter count.
- Keep it; Part 2 upgrades ARCH to GARCH.

Deliverable: `arch_lab.py` + a conditional-variance band over your returns.

Three Questions Before You Move On

If you can answer these, Part 1 has landed

Q1

Returns are uncorrelated but squared returns are not. What does this tell you?

ARCH effects present

Q2

What is the difference between conditional and unconditional variance?

Time-varying vs fixed

Q3

Why does pure ARCH need so many lags for real data?

Long memory in vol

All three come from the conditional-variance idea and the clustering engine.

Three Misconceptions to Avoid

Where volatility modeling trips people up

Common myths

- "ARCH lets me forecast the direction of returns."
- "Volatility clustering means returns are autocorrelated."
- "A constant variance is fine most of the time."

The reality

- ARCH forecasts the SIZE of moves, never their sign.
- Returns stay uncorrelated; it is their SQUARES that cluster.
- Constant variance is wrong exactly when it matters — in a crisis.

Three Things to Carry Forward

Part 1 in one screen

1

Variance is the target now, not the mean

Returns are unpredictable in direction but predictable in size. ARCH models the conditional variance — today's risk, given the recent past.

2

ARCH ties variance to recent squared shocks

A big shock raises tomorrow's variance, creating the self-reinforcing loop that produces clustering — and fat tails as a bonus.

3

ARCH works, but is inefficient

Matching volatility's long memory needs many ARCH lags. One extra idea — memory of past variance — fixes that. Enter GARCH.

Next — Part 2: GARCH, the workhorse that adds variance memory and rules the world's risk desks.